

Comparative Study of Post-Hoc Methods vs Intrinsic Interpretability in Vision Models

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The "Black Box" Problem

- Deep learning models (CNNs, ViTs) have opaque decision-making processes.
- **Post-Hoc XAI**: Explainers applied *after* training (e.g., **SHAP** [1], **LIME** [2]).
- **Intrinsic XAI**: Models interpretable by design (e.g., **ProtoPNet** [3]).

- **SHAP**

$$\phi_i = \sum_S \frac{|S|!(n-|S|-1)!}{n!} \Delta v$$

- **LIME**

$$\xi(x) = \arg \min_{g \in G} \mathcal{L} + \Omega(g)$$

- **GradCAM**

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$$

- **ProtoPNet**

$$\text{sim} = \log \left(\frac{d^2 + 1}{d^2 + \epsilon} \right)$$

- **Infidelity**

$$\text{INFD} = \mathbb{E}_I \left[(f(x) - f(x - I) - I^T \Phi(x))^2 \right]$$

- **Complexity (Entropy)**

$$C(\Phi) = - \sum_i p_i \ln(p_i)$$

- **Localization (IoU)**

$$\text{Loc} = \frac{|M \cap B|}{|M \cup B|}$$

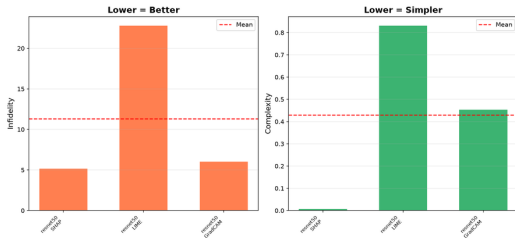
- **Rank Agreement (Spearman)**

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Experimental Setup & Metrics

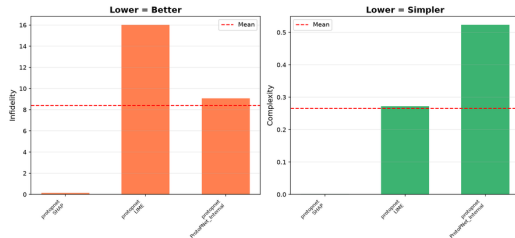
Post-Hoc Metrics

XAI Metrics Comparison



Intrinsic Metrics

XAI Metrics Comparison

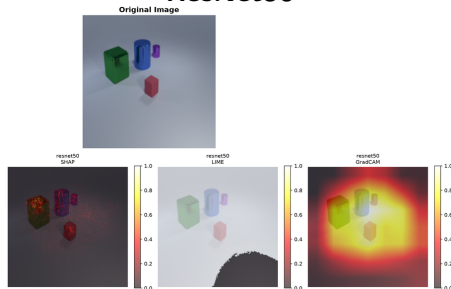


Datasets: *CLE4EVR* (3D objects) and *MNMath* (Handwritten equations).

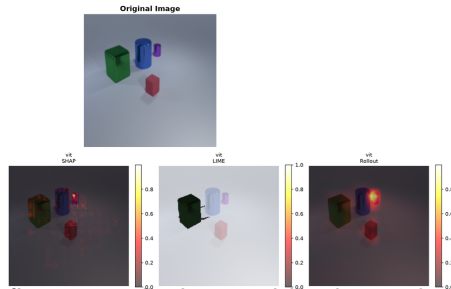
Key Metrics: *Infidelity*, *Complexity*, *Localization*.

Post-Hoc Methods on CLE4EVR Dataset

ResNet50



Vision Transformer (ViT)

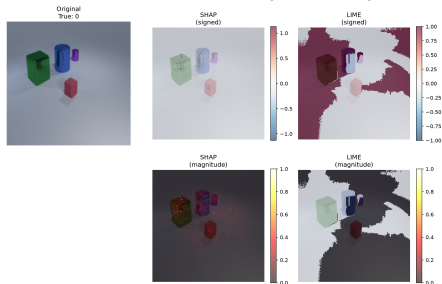


- **SHAP** achieves the lowest complexity, isolating fine-grained edges and bright pixels with high sparsity.
- **GradCAM** performs well: large 3D objects match its coarse receptive field.
- **LIME** underperforms: default SLIC superpixels severely fragment the smooth 3D shapes.

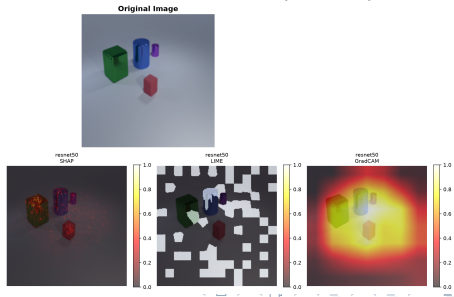
LIME Ablation Study on CLE4EVR

- **Hypothesis:** Does matching LIME's superpixels (K) to the number of objects improve explanations?
- **Result:** Object-aligned boundaries ($K = 4$) unexpectedly double the *Infidelity* compared to default ($K = 50$). Linear surrogates fail to capture the non-linear reasoning of CNNs.

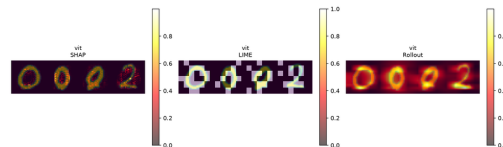
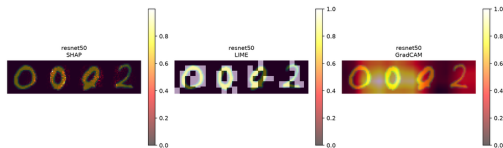
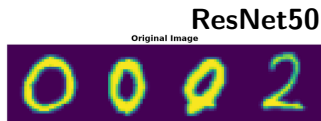
Default LIME ($K = 50$)



Ablated LIME ($K \approx 4$)



Post-Hoc Methods on MNMath Dataset

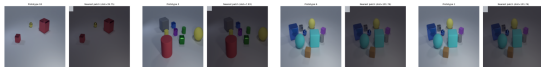


- **SHAP** heavily outperforms GradCAM on fine-grained stroke detection.
- GradCAM's low spatial resolution blurs over the thin handwritten digits.
- **LIME** struggles to isolate thin strokes, as superpixels tend to group the background with the fine handwritten details.

ProtoNet Learned Prototypes

- ProtoNet inherently learns a set of visual *prototypes* directly from the training data.
- Classification is based on the distance between the input image patches and these learned prototypes.

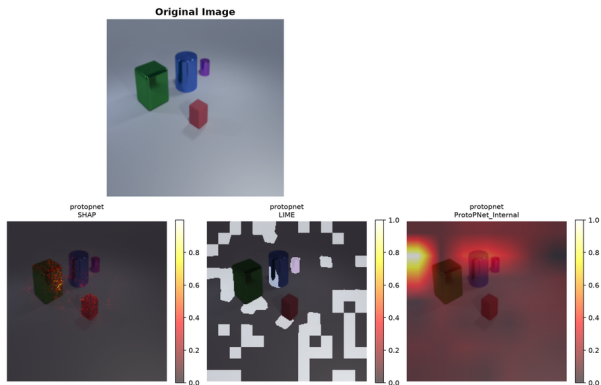
CLE4EVR Prototypes



MNMath Prototypes

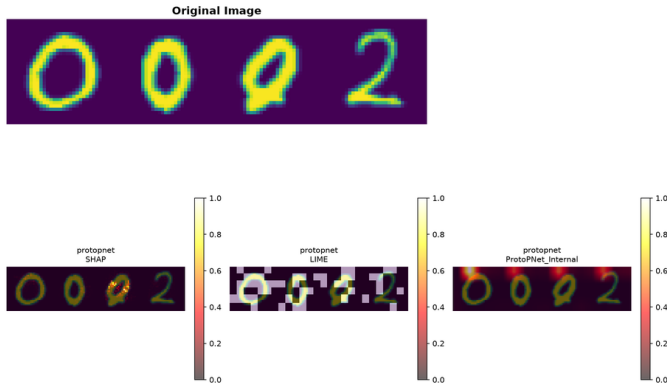


Visual Comparison on CLE4EVR



SHAP isolates edges (sparse), while **ProtoNet Internal** highlights broad, context-aware regions.

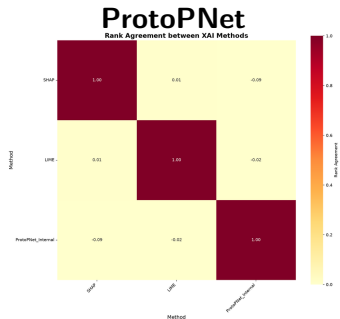
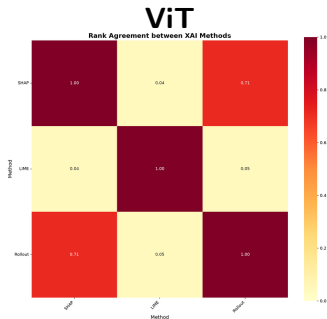
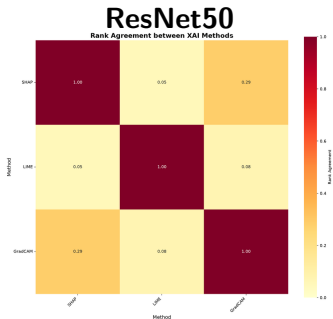
Visual Comparison on MNMath



ProtoPNet reasoning encompasses the whole digit and its context, completely ignoring the strict "stroke" pixels favored by SHAP.

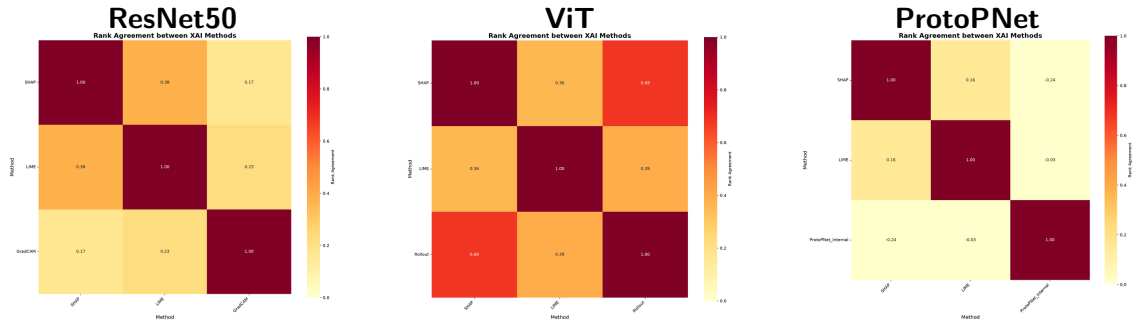
Rank Agreement: CLE4EVR Dataset

- Correlation between Spatial Locality (Intersection) and Infidelity.
- Across all architectures, post-hoc rank correlation is **near zero or negative**.



Rank Agreement: MNMATH Dataset

- The disconnect is even worse on fine-grained data (-0.24 for ProtoPNet).
- Post-hoc methods misrepresent the actual broad spatial reasoning of the network!



Key Takeaways

- Relying strictly on "Localization Accuracy" is dangerous: it assumes networks reason using human-defined object boundaries.
- Intrinsic interpretability reveals that models learn broad spatial contexts, unlike what post-hoc methods suggest [4].

Thank You!

Questions?

References I

-  S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” *Advances in neural information processing systems*, vol. 30, 2017.
-  M. T. Ribeiro, S. Singh, and C. Guestrin, “‘why should i trust you?’ explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
-  C. Chen, O. Li, D. Tao, A. Barnett, C. Rudin, and J. K. Su, “This looks like that: deep learning for interpretable image recognition,” in *Advances in neural information processing systems*, vol. 32, 2019.
-  C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.